

Pass-Through, Welfare, and Incidence under Product Returns

Draft notes

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Abstract

With product returns, local welfare and incidence weight kept units rather than gross sales, and pass-through is unchanged once list price is replaced by effective margin.

1 Setting

A homogeneous product is sold at list price p . Consumers indexed by i differ in return hassle κ_i , the disutility from returning the product, and in match distribution G_i over post-purchase match quality u . After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u \sim G_i} [\max(u - p, -\kappa_i)].$$

For purchase, all consumer-side heterogeneity summarizes into willingness to pay v_i , defined by $s_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, $F(v) = \Pr(v_i \leq v)$, gross demand is $D(p) = 1 - F(p)$, and share $r(p)$ of gross units sold at p are returned.

The product market is perfectly competitive, producer j takes p as given, chooses output at cost $C_j(q_j)$, and pays handling cost $h > 0$ on each returned unit.

2 Consumer side

Write $s(v, p)$ for expected surplus at list price p among types with willingness to pay v , so $s(v, v) = 0$. Consumer surplus integrates surplus over all types who buy:

$$CS(p) = \int_p^\infty s(v, p) dF(v). \quad (1)$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers:

$$CS'(p) = -s(p, p) f(p) + \int_p^\infty s_p(v, p) dF(v),$$

where $s_p(v, p) \equiv \partial s(v, p) / \partial p$ and f is the density of v_i when F is differentiable. On the extensive margin, $s(p, p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v, p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase:

$$s_p(v, p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty s_p(v, p) dF(v) = -D(p)(1 - r(p)),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)(1 - r(p)) dp. \quad (2)$$

The extensive- and intensive-margin decomposition makes the logic transparent, but the local surplus loss depends only on the aggregates $D(p)$ and $r(p)$. Price enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet any microstructure that delivers the same gross demand $D(p)$ and return share $r(p)$ generates the same dCS . In particular, it does not matter whether high- v_i types adjust return behavior more steeply than low- v_i types when p changes: only kept units enter at first order, through the factor $1 - r(p)$, not gross purchases $D(p)$ alone.

3 Supply side

Returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus handling h if the item is returned. Among gross sales at list price p , share $r(p)$ are returned, so expected revenue per gross unit—the *effective margin*—is

$$\mu(p) \equiv p - r(p)(p + h). \quad (3)$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p)$.

Given μ , firm j chooses output to solve

$$\max_{q_j \geq 0} \mu q_j - C_j(q_j), \quad \text{FOC: } C'_j(q_j) = \mu \text{ (if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j 's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit:

$$S(\mu) \equiv \sum_j q_j(\mu), \quad PS(\mu) \equiv \sum_j [\mu q_j(\mu) - C_j(q_j(\mu))]. \quad (4)$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu} [\mu q_j - C_j(q_j)] = q_j + q'_j(\mu) [\mu - C'_j(q_j)].$$

Active firms satisfy $C'_j(q_j) = \mu$ at their chosen output, so the adjustment term $q'_j(\mu) [\mu - C'_j(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve:

$$PS(\mu) = \int_0^\mu S(x) dx. \quad (5)$$

At equilibrium $\mu = \mu(p)$, $PS(\mu(p)) = \int_0^{\mu(p)} S(x) dx$.

4 Pass-through

Equilibrium clears the market when gross demand equals supply at the effective margin implied by returns, $D(p) = S(\mu(p))$ with $Q \equiv D(p)$. Consider a small cost shock x , such as a uniform upward shift in every C'_j or an increase in handling cost h , and write $\rho_x \equiv dp/dx$. Let $m(p)$ denote the marginal active firm's per-unit margin. Perfect competition gives $m(p) = 0$ in equilibrium. Write $m_p \equiv dm/dp$ and $m_x \equiv \partial m/\partial x|_{Q,p}$. Holding (p, Q) fixed, a uniform production-cost shift gives $m_x = -1$ and a handling shock gives $m_x = -r(p)$. Differentiating $D(p) = S(\mu(p))$ and $m(p) = 0$ along the shocked equilibrium path then gives

$$\rho_x = \frac{S'(\mu) m_x}{S'(\mu) m_p - D'(p)}, \quad \rho_h = r(p) \rho_C. \quad (6)$$

Here $S'(\mu) = \sum_j q'_j(\mu)$ is the aggregated slope of firm supply. No single aggregate $C(Q)$ is needed. In the homogeneous special case ($C_j = C$), $m_p = 1 - C'''(Q)D'(p) - r'(p)(p + h) - r(p)$ and the formula above reduces to the standard competitive pass-through when $r \equiv 0$.

5 Incidence

Combining $dCS = -D(p)(1 - r(p)) dp$ with the supply-side objects above, the welfare incidence of the shock is

$$\frac{dCS}{dx} = -D(p)(1 - r(p)) \rho_x, \quad \frac{dPS}{dx} = Q(m_p \rho_x + m_x). \quad (7)$$

The incidence ratio $I_x \equiv |dCS/dx|/|dPS/dx|$ is

$$I_C = I_h = \frac{(1 - r(p)) S'(\mu)}{-D'(p)}. \quad (8)$$

Production and handling shocks bear the same consumer share because $\rho_h = r(p) \rho_C$ and the factor $1 - r(p)$ applies to both.